



Persuasive Technology

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What is persuasion

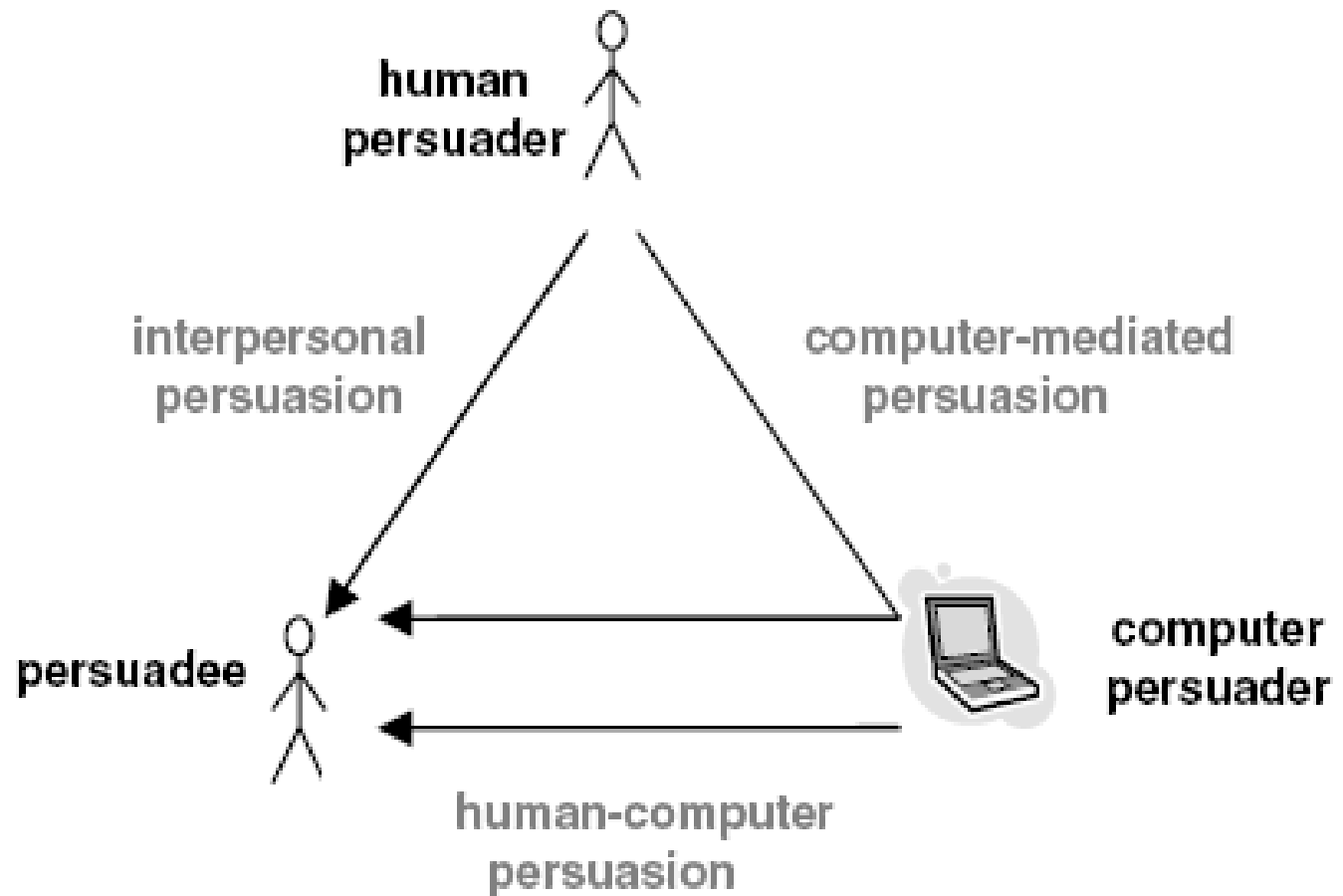
Persuasion



- Influencing attitudes and behavior through communication



Persuasion



Interpersonal persuasion

- *Interpersonal persuasion occurs when two or more people interact with each other.*
- Interpersonal communication involves **verbal** and **non-verbal forms** of behavior,
- personal feedback, coherence of behavior and an intent to change the attitudes and/or behavior of other(s)



Computer-mediated Persuasion.

- People may persuade also others through e-mail, instant messages, or other
- computer technologies, i.e. through computer-mediated communication

Human-computer persuasion

- *differs from other persuasion types in that it is not always clear who is the persuader. As computers do not have intentions of their own, those who create, distribute, or adopt the technology have the intention to affect someone's attitudes or behavior. These intentions can be described as endogenous, exogenous or autogenous*

Persuasion Variables (Who)

Source:

- attraction,
- similarity,
- trustworthiness,
- expertise,
- power, status,
- communication style
 - (appearance, voice, gestures,),
 - individual vs group;
 - single vs multiple.

Persuasion Variables (What)

Message:

- argument strength(newness, valence, trueness),
- discrepancy,
- emotional vs cognitive;
- positive vs negative(fear)
- appeals,
- humour,
- type of language(eg metaphors, complexity, concrete vs abstract), one sided vs two sided, implicit vs explicit conclusions, including vs omitting weak arguments; message length. overload, repetition

Persuasion Variables (How)

Channel:

- sensory mode,
- verbal/nonverbal,
- physical/ electronical media,
- interactivity& intelligence,
- personal vs mass media.

Persuasion Variables (To whom)

Receiver:

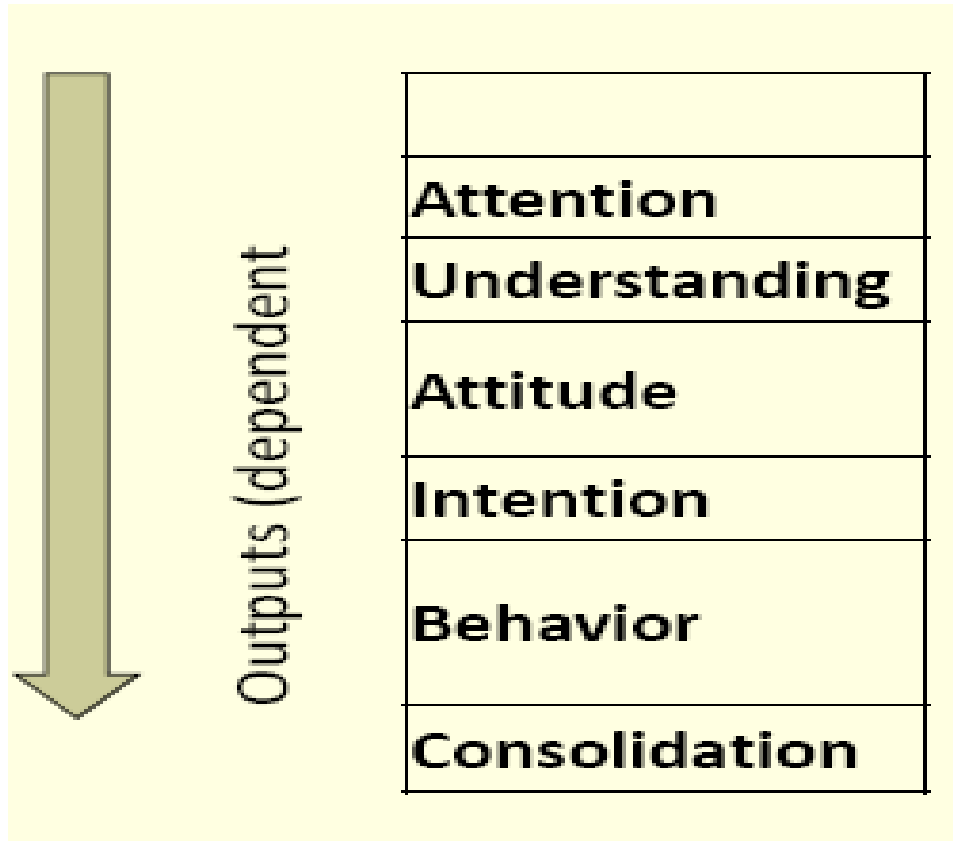
- prior attitude,
- attitude function,
- confidence, ambivalence, involvement, commitment,
- self-esteem,
- active participation, intelligence, prior beliefs and knowledge, other attitudes, etc.

Persuasion Variables (Where)

Context:

- Group context ;
- Individual context ;
- Noisy vs. quiet environment; etc.

Persuasion Phases



Steps that individuals must be persuaded to take for message to have impact



What is Persuasive Technology

Persuasive Design?

- **Usability focuses** on giving users the ability to complete a transaction if they are desire.
- A usable site makes it easy for users to complete transactions, from buying products to convincing users to read featured articles.
- To be successful, sites must go beyond Usability by focusing on Persuasive Design. They **must motivate users** by taking advantage of persuasive tactics that will make them take action. The most persuasive web sites focus on making users feel comfortable about making decisions and helping them act on them.



What is Persuasive Technology ?

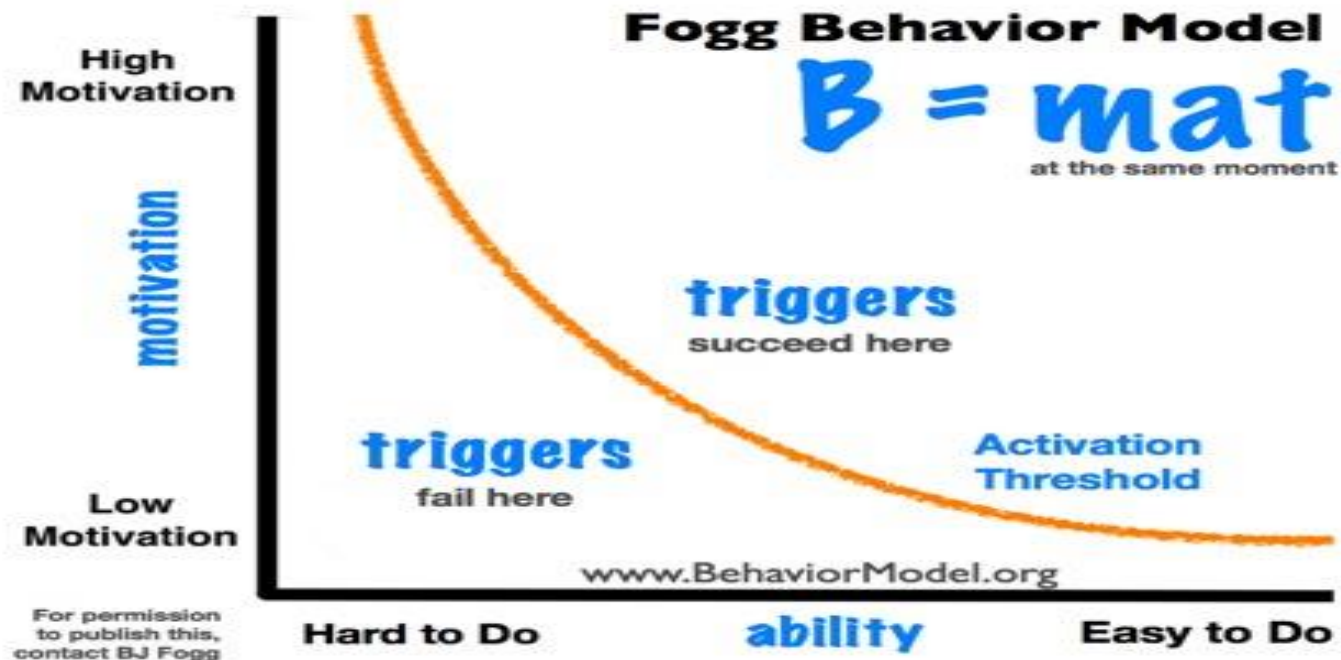
- Persuasive technology is any interactive computing system designed to change people's attitudes or behaviors(Fogg, 2003).
- A class of technologies that are intentionally designed to change a person's attitude or behaviour (Ijsselsteinet al., 2006).

Persuasive Technology

- Intentional attempt to change attitudes or behaviors
- non-coercive or deceptive; the grey area of pre-conscious persuasion.
- computer as change agent interaction with vs through computers
- however: still the full input-output matrix applies: eg no persuasion without attention, or attitude retrieval etc.

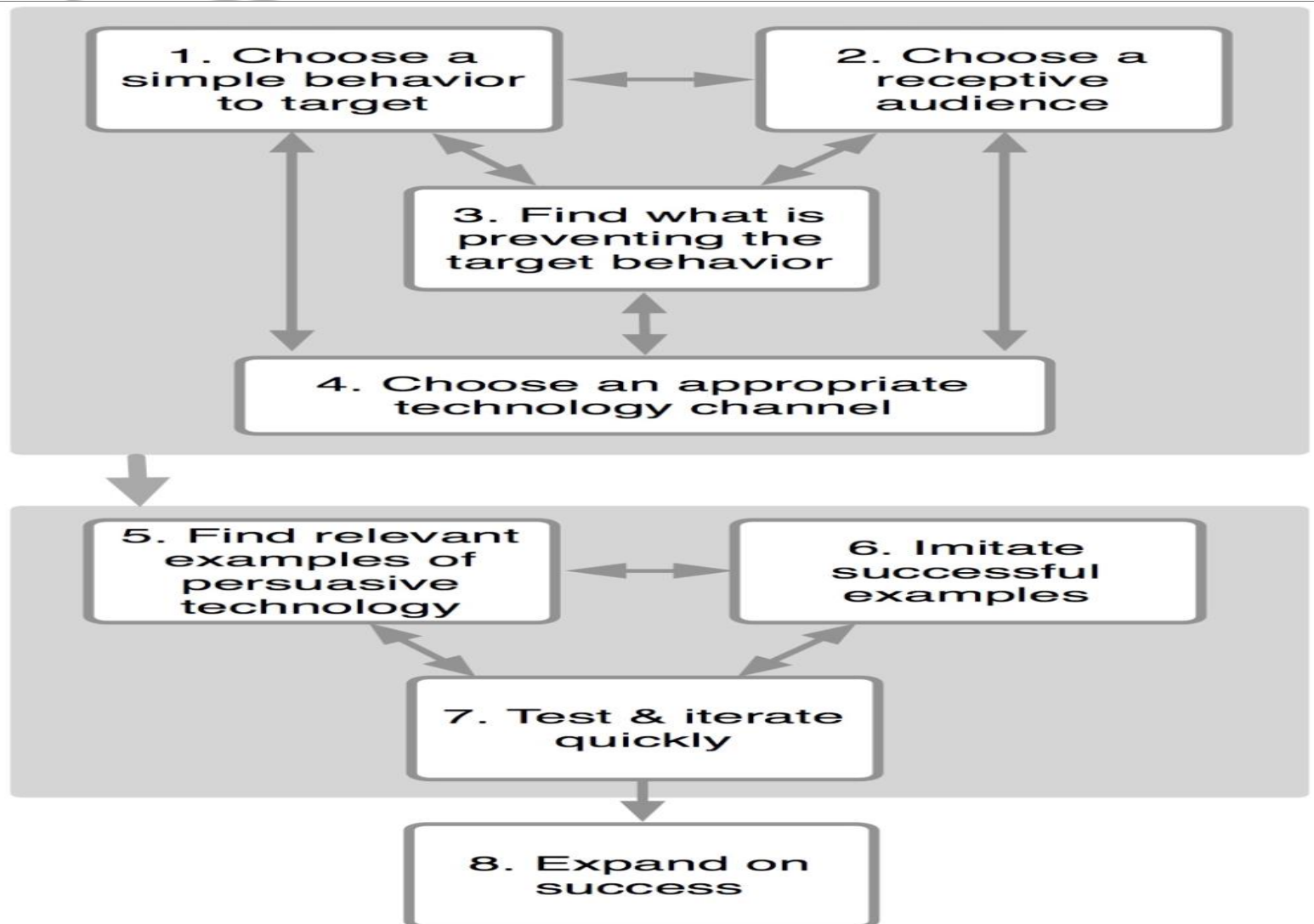
What Causes Behavior Change?

- My Behavior Model shows that **three elements must converge at the same moment** for a behavior to occur: **Motivation**, **Ability**, and **Trigger**. When a behavior does not occur, at least one of those three elements is missing.



Eight-Step Design Process

Bj Fogg



Before



After

Behavior Change Elements
motivation, ability, trigger

Core Motivators
pleasure/pain, hope/fear,
social acceptance/rejection

Simplicity Factors
time, money, physical effort,
brain cycles, social deviance,
non-routine

Triggers
facilitator, spark, signal

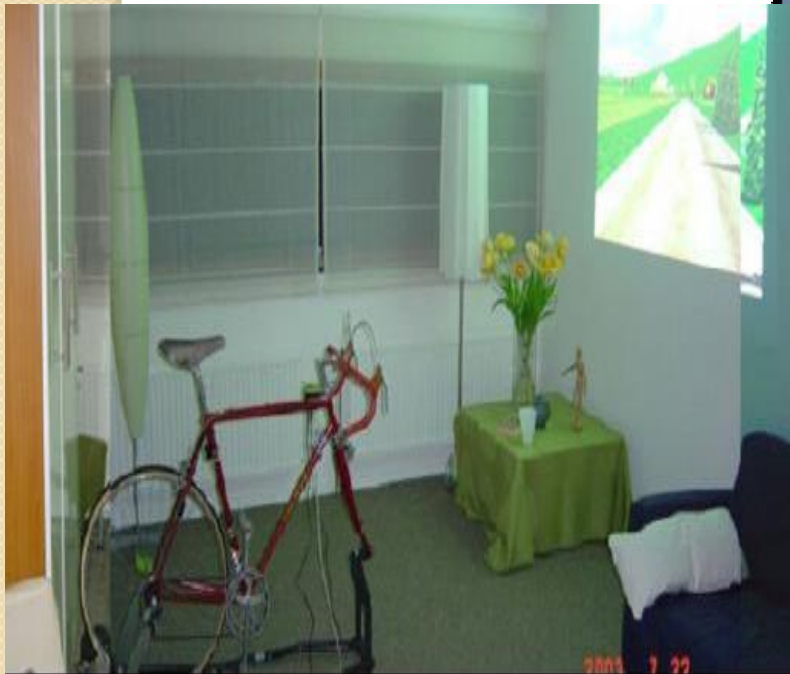




Examples

Virtual Coach

- **Effects of Immersion and a Virtual Coach on Motivation and Presence with a Home Fitness Application**



Persuasive Gaming

- Creates rewarding situations
- learning about world
- Ideological persuasion > *ethical issues*



Persuasive Multimedia Learning



This persuasive multimedia learning environment (PMLE) attempted to persuade children to reduce their dental anxiety

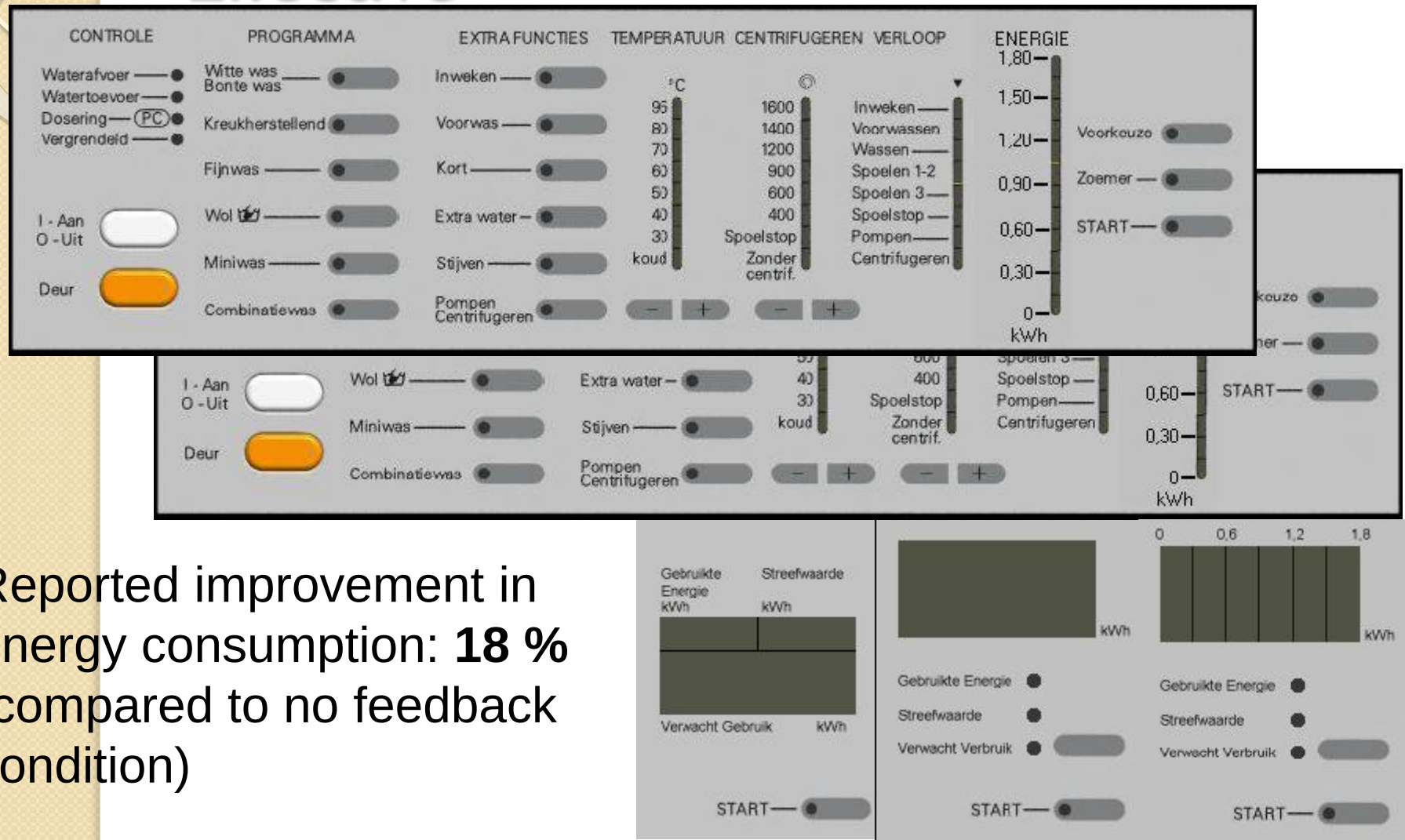


Illustrations in energy consumption

- blue light signals outdoor temperature for switching off air co(Seligman, 1978)> *prompt*
- monitor indicates electricity consumption rate/costs (eg Hutton) >aggregated *behavioral feedback*
- thermo-sensitive tiles in bathroom: *non-verbal behavioral feedback*
- visualized electric pulses: *issue attention enhancement*
- monitor gives interactive choice feedback before overt action (McCalley, 2002): *action specific interactive feedback*



Interactive Feedback can be Effective



Persuasive Mirror

- Visual behavioral feedback for experiencing highly self-relevant effects of unhealthy behavior



Experiencing Future Risks

- Lack of awareness and direct experience may be compensated by simulated experience



Custard puddings produced in the Polite condition (left) and in the Casual condition (right).



Virtual Treatment of Phobias and Fears

- Phobia treatment using virtual environments
- Systematic desensitization comparable to standard procedure
- Generalization to real world

e.g, Fear of flying

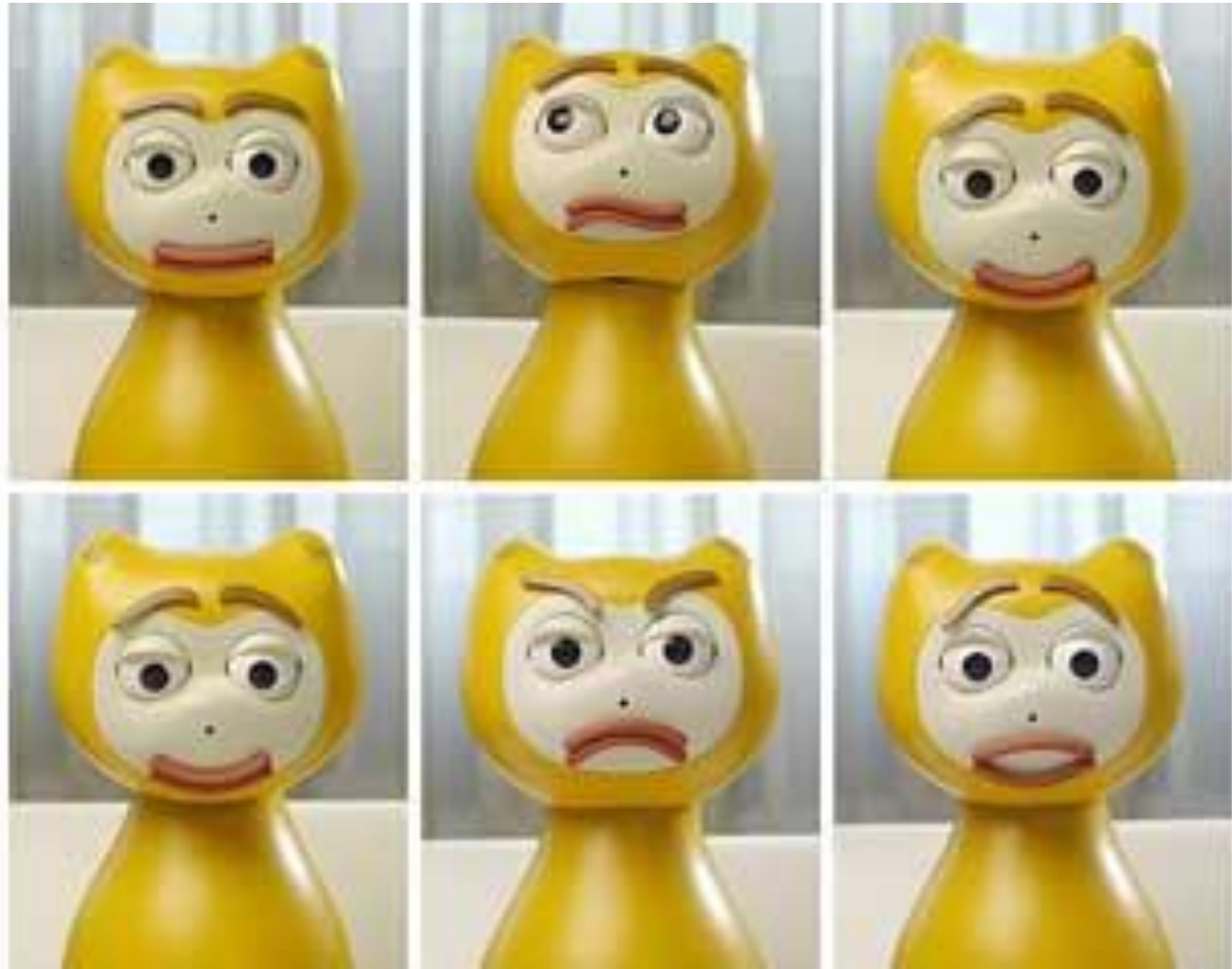


Systems as Persuasive Social Actors

- Rewarding people
- Modeling behavior or attitude
- Providing social support

Can a system be social and what makes it social?

Social Actor?



Persuasive Social Actors

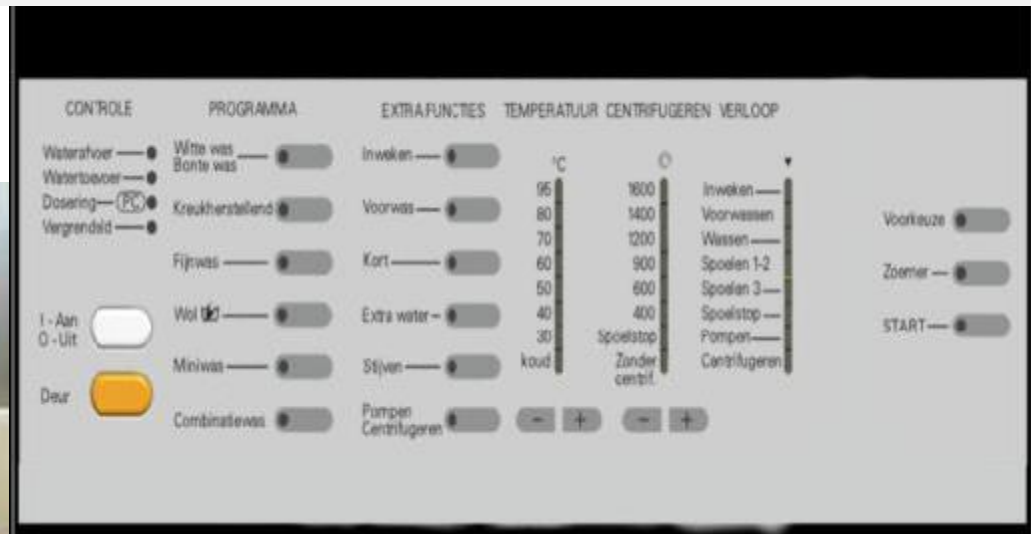


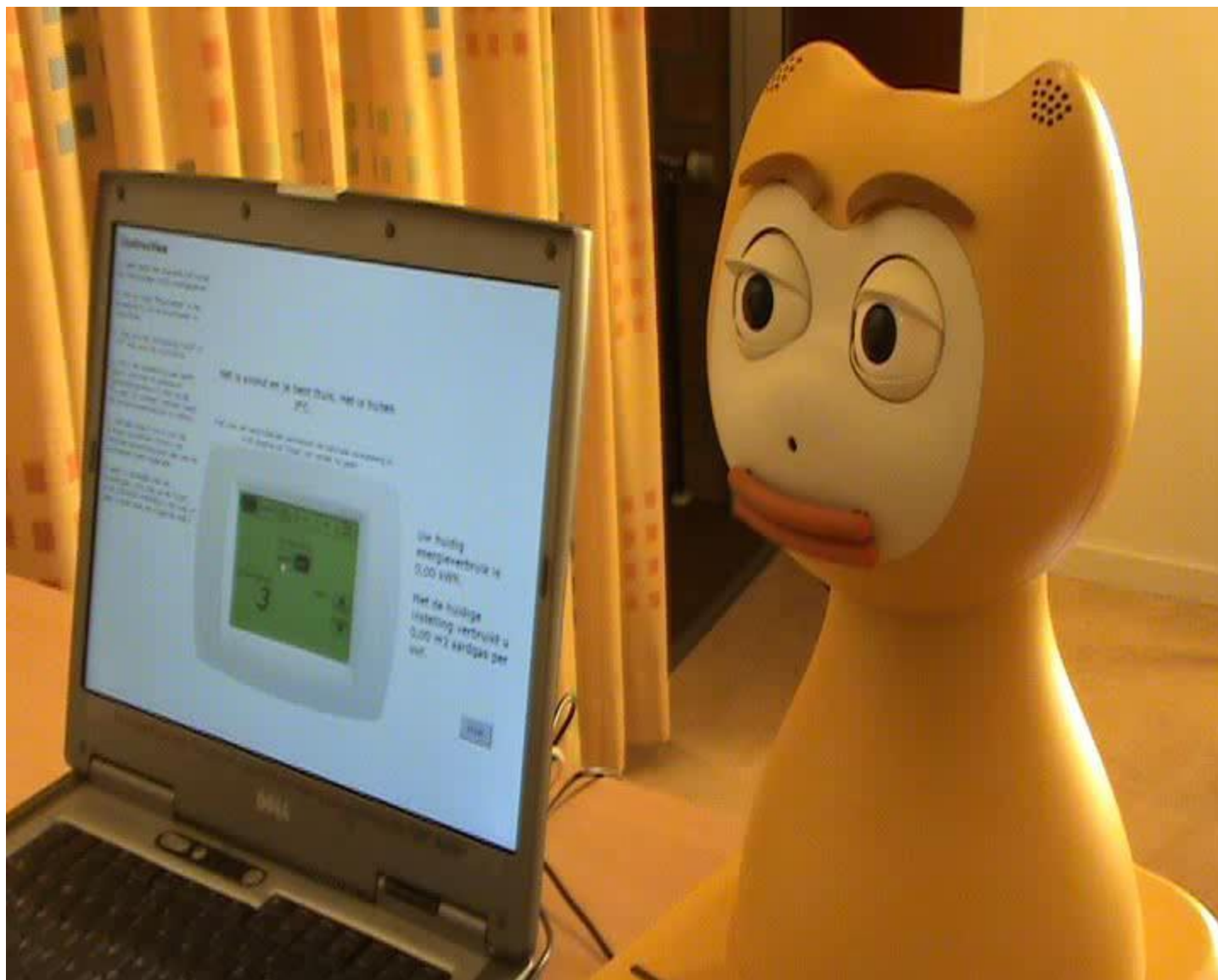
Social Feedback

Feedback condition: High-agency social feedback

This is Victor, an advanced robot with a mind of his own. He will provide you with feedback on your electricity use.

You have to
set the
temperature
to 60°C





An Example of a Good Site Persuading Users

- eBags.com is one of the most persuasive sites I've seen. The site's designers motivate users to purchase bags by offering them detailed content that helps them pick the right bag for their specific needs.

Search eBags for 
[Detailed Search](#) | [Shop Our Catalog](#) | [Request A Catalog](#)

Get **FREE SHIPPING** on your entire order! Also, shop our [Winter Clearance Sale](#) for items already up to 60% off, U.P.S. Ground shipping only, no AK or HI. Cannot be combined with other offers.

[Return to Product Detail Page](#)
Model: A.T. Gear Deluxe Computer Case**Brand:** High Sierra**MSRP:** \$120.00**Our price:** \$79.99 (33% off)**Please Select a Color**

2 Nightfall/Black



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Site's Biggest Strengths

- First, they provide multiple shots of the product from various angles so that users can really see what the bag looks like.
- Second, the pictures show the bags stuffed with the items that are likely to go in them, displaying pens, mobile phones, notebooks, and laptop computers to give users a sense of how much stuff the bag can hold..



Persuasive Patterns

1. Cognitive
2. Game Mechanics
3. Perception and memory
4. Feedback
5. Social

- <http://ui-patterns.com/patterns>

I.Cognition

Patterns of psychological tendencies that cause the human brain to draw incorrect conclusions

- Loss Aversion
- Scarcity
- Other cognitive biases

<http://ui-patterns.com/patterns/cognition/list>

2. Game mechanics

Games engage, involve, and influence us through its playful nature.

- Gameplay design
- Fundamentals of rewards
- Game
- play rewards

<http://ui-patterns.com/patterns/game-mechanics/list>

3. Perception and memory

Patterns in how we visually perceive, interpret, and remember meanings as we interact with systems.

- Attention
- Comprehension

<http://ui-patterns.com/patterns/perception-and-memory/list>

4. Feedback

As the users interacts with your system feedback motivate them to take the next step.

- Timing

<http://ui-patterns.com/patterns/feedback/list>

5. Social

Allow the user to associate, communicate, and interact with other people online.

- Reputation
- Social interactions
- <http://ui-patterns.com/patterns/social/list>

Disadvantages of traditional PT

- Traditional PT
 - needs focal attention
 - takes away cognitive resources
 - users must be motivated to pay attention to it
- A challenge to traditional Persuasive Technology:
 - In real-life situations, people do not have these cognitive resources available!
 - Persuasive technology might even give rise to reactance

Subliminal Feedback

- feedback for 22 ms.
 - Supraliminal (interactive) feedback and subliminal (interactive) feedback both led to more correct energy consumption ratings as compared to receiving no feedback
- It can influence **unconsciously!**

Per Frames: Persuasive Picture Frames for Proper Posture



Sources

- Jaap Ham, Human Technology Interaction
- Eindhoven University of Technology
- B. J. Fogg. Persuasive Technology: Using Computers to Change What We Think and Do. Morgan Kaufmann, San Francisco, 2003.